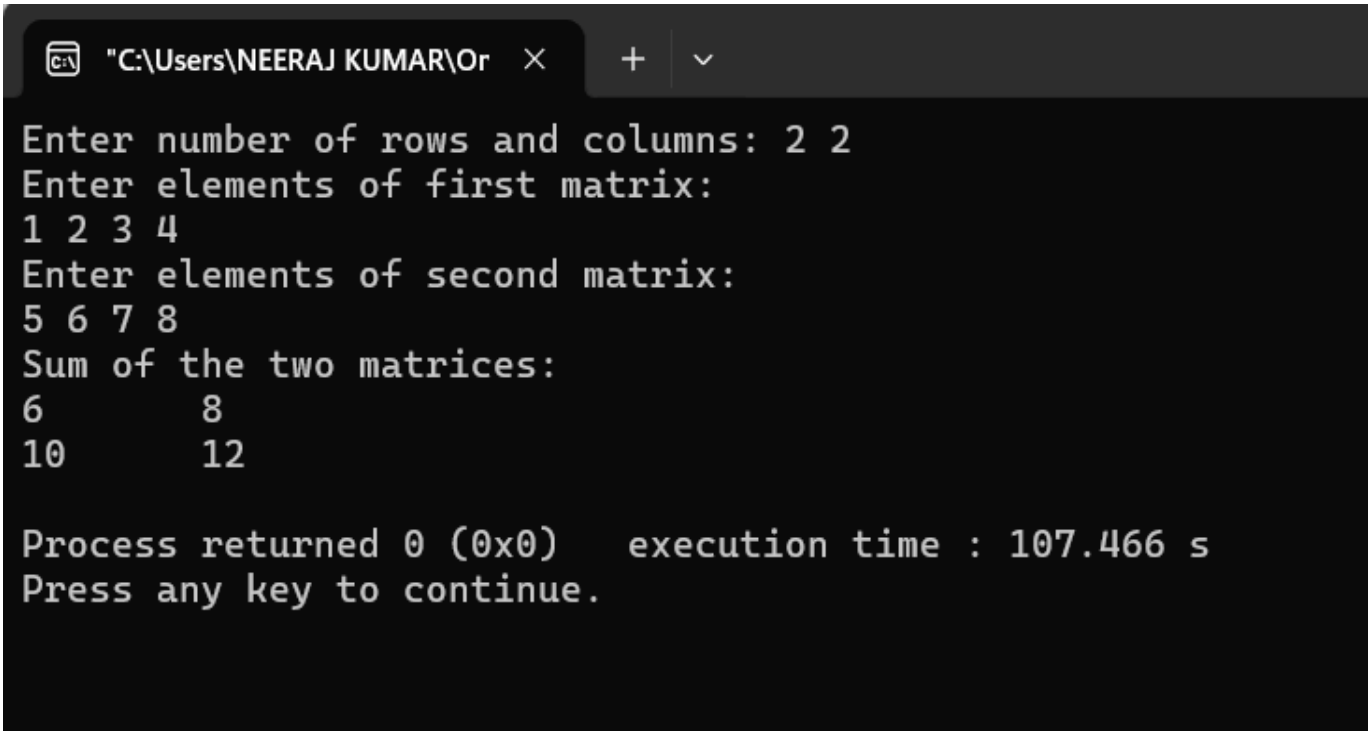
Program :

```
#include <stdio.h>
int main() {
int r, c, i, j;
printf("Enter number of rows and columns: ");
scanf("%d %d", &r, &c);
int a[r][c], b[r][c], sum[r][c];
printf("Enter elements of first matrix:\n");
for(i = 0; i < r; i++)
for(j = 0; j < c; j++)
scanf("%d", &a[i][j]);
printf("Enter elements of second matrix:\n");
for(i = 0; i < r; i++)
for(j = 0; j < c; j++)
scanf("%d", &b[i][j]);
// Add corresponding elements
for(i = 0; i < r; i++)
for(j = 0; j < c; j++)
sum[i][j] = a[i][j] + b[i][j];
printf("Sum of the two matrices:\n");
for(i = 0; i < r; i++) {
for(j = 0; j < c; j++)
printf("%d \t", sum[i][j]);
printf("\n");
}
return 0;
}
```

Output :